FINITE PUSHFORWARDS KILL HIGHER COHOMOLOGY AND PRESERVE COHERENCE: FOR F : X -> Y FINITE WITH Y LOCALLY NOETHERIAN AND F QUASI-COHERENT, R^PF_*F = 0 FOR P > 0, AND F_*F IS COHERENT WHENEVER F IS.

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2671-2678 of sources/stacks-coherent/pinned-original-source.tex; proof span is lines 2680-2695, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool stacks-coherent backfill wave 33f.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; every pointer named here occurs verbatim in the retained statement/proof text.

Source reference. Original lemma 00GJ: sources/stacks-coherent/pinned-original-source.tex, source lines 2695–2695. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01XZ: sources/stacks-coherent/pinned-original-source.tex, source lines 2442–2455. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01XC: sources/stacks-coherent/pinned-original-source.tex, source lines 398–402. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01WH: sources/stacks-coherent/pinned-original-source.tex, source lines 2695–2695. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01T6: sources/stacks-coherent/pinned-original-source.tex, source lines 2695–2695. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:coherent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Assume f is finite and Y locally Noetherian. Then $R^p f_* \mathcal{F} = 0$ for $p > 0$ and $f_* \mathcal{F}$ is coherent if $\mathcal{F}$ is coherent.*

Proof. The higher direct images vanish by Lemma 01XC and because a finite morphism is affine (by definition). Note that the assumptions imply that also X is locally Noetherian (see Morphisms, Lemma 01T6) and hence the statement makes sense. Let $\mathrm{Spec}(A) = V \subset Y$ be an affine open subset. By Morphisms, Definition 01WH we see that $f^{-1}(V) = \mathrm{Spec}(B)$ with $A \to B$ finite. Lemma 01XZ turns the statement of the lemma into the following algebra fact: If M is a finite B-module, then M is also finite viewed as a A-module, see Algebra, Lemma 00GJ. □